

Time : 3 Hrs.]

[Max. Marks :70

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

Note: Attempt all questions. All questions carry equal marks. Each question carry five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

		Marks	CO	BL	PI
Q.1	(a) Write the contribution in mathematics by Varahamihira.	02	1	1	1.1.1
	(b) If $u = x^2(y-z) + y^2(z-x) + z^2(x-y)$, then find $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.	02	1	1	1.1.1
	(c) If $y_1 = \frac{x_2 x_3}{x_1}, y_2 = \frac{x_3 x_1}{x_2}, y_3 = \frac{x_1 x_2}{x_3}$, then find $\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)}$.	03	1	2	1.1.1
	(d) Expand $\cot^{-1}(xy)$ in powers of $(x+0.5)$ and $(y-2)$ up to second degree terms. Hence compute $f(-0.4, 2.2)$ approximately.	07	1	2	2.1.3
	OR				
	(e) Examine the function $f(x, y) = y^2 + 4xy + 3x^2 + x^3$ for maximum or minimum values. Also find the function value at extreme point.	07	1	3	2.1.3
Q.2	(a) Find the value of $\int_0^{\infty} \frac{e^{-st}}{\sqrt{t}} dt$.	02	2	1	2.1.3
	(b) Evaluate $\int_0^1 x^2 (1-x)^3 dx$ using beta function.	02	2	1	2.1.3
	(c) Evaluate $\int_0^1 \frac{x^4 (1+x^5)}{(1+x)^{15}} dx$.	03	2	2	2.1.3
	(d) Evaluate the integral $\iint \sqrt{a^2 - x^2 - y^2} dx dy$ over the semicircle $x^2 + y^2 = ax$ in the positive quadrant by changing into polar coordinates.	07	2	3	2.1.3
	OR				
	(e) Change the order of integration in $\int_0^1 \int_y^1 x^2 \cos(x^2 - xy) dx dy$ and hence evaluate it.	07	2	2	2.1.3
Q.3	(a) Show that every square matrix is expressible as the sum of a Hermitian matrix and a skew-Hermitian matrix.	02	3	1	1.1.1
	(b) Define the rank of matrix with example.	02	3	2	1.1.1
	(c) Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$	03	3	2	1.1.1

- (d) Investigate for consistency of the following equations and if possible, find the solutions. 07 3 2
- $$2x_1 + x_2 + 2x_3 + x_4 = 6, 6x_1 - 6x_2 + 6x_3 + 12x_4 = 36,$$
- $$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1, 2x_1 + 2x_2 - x_3 + x_4 = 10$$

OR

- (e) Verify Cayley-Hamilton theorem for the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

07 3 3 1

- Q.4 (a) Define linear differential equation with example. 02 4 1 2

(b) Solve $(3x+2)^2 \frac{d^2 y}{dx^2} + 3(3x+2) \frac{dy}{dx} - 36y = 0$ 02 4 1 2

- (c) Find the particular integral of $(D^3 + 1)y = \sin(2x + 3)$ 03 4 2 2

- (d) Apply the method of variation of parameters to solve 07 4 3 2

$$\frac{d^2 y}{dx^2} + 4y = \cot 2x$$

OR

- (e) Uranium disintegrates at a rate proportional to the amount present at any instant. If M_1 and M_2 grams of uranium are present at times T_1 and T_2 respectively, Show that the half-life of uranium is $\frac{(T_2 - T_1) \log 2}{\log \left(\frac{M_1}{M_2} \right)}$. 07 4 2 2

- Q.5 (a) Define core of a membership function and decomposition of a fuzzy set with suitable example. 02 5 2 1

- (b) Define triangular and trapezoidal membership function. 02 5 2 1

- (c) Prove that a fuzzy set A on R is convex if and only if $A(\lambda x_1 + (1-\lambda)x_2) \geq \min[A(x_1), A(x_2)]$ for all $x_1, x_2 \in R, \lambda \in [0, 1]$. 03 5 2 1

- (d) If fuzzy sets A and B defined on X such that 07 5 3 1.1

$$A = \{(1, 0.2), (2, 0.4), (3, 0.6), (4, 0.8), (5, 1)\},$$

$$B = \{(1, 0.4), (2, 0.3), (3, 0.2), (4, 0.9), (5, 0.7)\},$$

then find product of A and B , difference of A and B , disjunctive sum of A and B .

OR

- (e) If fuzzy sets A and B defined on X such that 07 5 3 1.1

$$A = \{(1, 0.3), (2, 0.5), (3, 0.8), (4, 1)\}$$

$$B = \{(1, 0.2), (2, 0.4), (3, 0.7), (4, 0.8)\}$$

then find $A \cap B$, relative cardinality of A and B , strong

α -cut set for $\alpha = 0.2$, power of fuzzy sets A and B for $\alpha = 3$.

